NAPROCK PROCON 2026 · Themed Section

BandFlow

Break it down, wear the flow.

AI-assisted work breakdown & smart-wristband task delivery



● Work in progress: voice entry and vibration alerts are in hardware assembly; benefits are proposed and not yet measured.

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1 The problem: too many open tasks

For everyone

Update campaign report	3
Reply to supplier thread	12
… 20+ more open items	

“What should I do next?”

Decision fatigue · Context switching · Dropped dependencies · Lower velocity

Answering this again and again costs time and energy. It is structural, not a personal failure.

2 Our idea: one step at a time

Focus on doing, not deciding

1 big task → Broken into ordered steps → Current step on the wrist

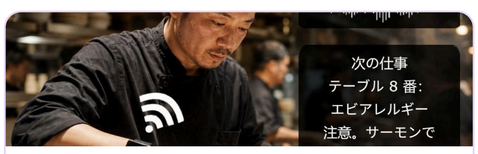
	Typical task tools	BandFlow
Worker sees	Whole backlog, alerts	**The current step**
Who decides next	The worker, again	**The ordered queue**
Where tasks arrive	Screens only	**Wristband, app, web**

3 Who it is for

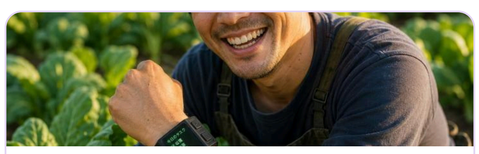
▲ *Photos and drawings are scenario illustrations, not real deployments.*



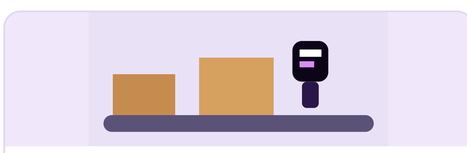
Hospital nurse
Moves from room to room
Wristband
Next task arrives as each is done



Restaurant kitchen
Hands full, noisy
Wristband
Vibration alert, tap to confirm



Farm / outdoor
No computer during work
Wristband · Web at home
Review progress at home



Factory / warehouse
Hands always occupied
Wristband
Tap DONE to finish the step



Office worker
Task pile-up
App / web
Priority matrix + ordered steps; no band needed

4 How it works

The core loop

1	2	3	4	5	6
SUPERVISOR	AI / SERVER	SERVER	WRISTBAND	WORKER	SERVER
Assign	**Break down**	**Queue**	**Deliver**	**Confirm**	**Advance**
Creates work in the app with priority and due date, picks the worker(s)	Ordered steps with prerequisites; the server validates them	First ready step becomes active; dependent steps stay locked	Sent over Bluetooth to that worker’s linked band; it vibrates and shows the step	Taps DONE on the band, or ticks the step in the app	Saves progress, unlocks and sends the next step, updates the dashboard

Steps 4 to 6 repeat until all steps are done. Example: “Launch a new seasonal menu”

1 List seasonal ingredients → 2 Draft 5 new dish ideas → 3 Cook and taste-test each dish → 4 Choose the final 3 dishes → 5 Calculate cost per dish → 6 Design and print the menu

5 System architecture

For specialists

AI service (optional) — Steps + priority class, as JSON
Whisper speech-to-text API — Voice clip in, text out
← internet; called only when needed

Mobile app — Expo / React Native
Web dashboard — React + Vite · same API

⇄ HTTP

Raspberry Pi 4 · local hub on the Wi-Fi network

- **Node.js API** — Accounts, work, steps, progress, pairing, voice
- **SQLite** — One source of truth: tasks, steps, events
- **Flask BLE bridge** — Bleak · auto-reconnect · event queue

Node → bridge: active step · Bridge → Node: DONE events and band requests

⇄ BLE

ESP32-S3 wristband — Touchscreen · vibration motor · electret mic · battery + charger

```
{ "title": "…", "subtasks": [
  { "description": "…",
    "order_index": 1,
    "depends_on_order_index": null } ] }
```

AI reply is validated by the server before saving (invalid steps and circular dependencies are rejected). **Priority:** AI classifier, or rules: urgent = due within 2 days, important = above Low priority.

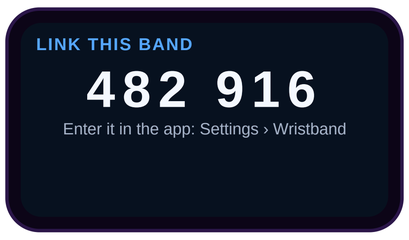
TypeScript · React Native · Expo · React · Vite · Node.js · SQLite · Python · Flask · Bleak · Arduino C++ · LVGL · NimBLE · ESP32-S3 · Bluetooth LE · Whisper API

AI and Whisper need internet; the rest runs on the local network.

6 The wristband

Hardware

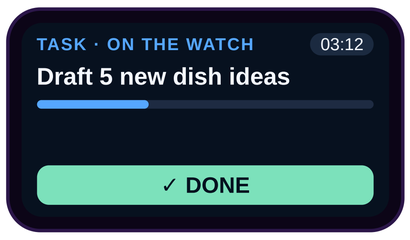
- **ESP32-S3 + touch** — 320×170 screen, DONE and menus
- **Electret mic** — Records voice for task entry
- **Vibration motor** — Alerts a new step
- **Battery + charger** — Rechargeable, charger module
- **Bluetooth LE** — Link to the Pi bridge
- **Account link** — 6-digit code, one band per worker



Pairing



Matrix



Current step

Screens drawn from the firmware design (example numbers). The band can also remove a step and unlink itself.

7 Voice task entry

Final hardware stage

1. **Speak** — Electret microphone on the ESP32-S3 records a short clip
2. **Send** — The clip travels over Bluetooth to the Raspberry Pi
3. **Transcribe** — The Pi sends the audio to the Whisper API and gets text back
4. **Confirm** — The text appears on the band’s screen to check
5. **Create** — Node saves the task, steps are built, and the first one is dispatched

8 Priority matrix

App + band

	URGENT	NOT URGENT
IMPORTANT	**DO FIRST** — Due within 2 days or overdue, priority above Low	**SCHEDULE** — Important, not due soon
MINOR	**DELEGATE** — Urgent, low priority	**ELIMINATE** — Low priority, not urgent

AI classifier when connected, else these rules. Band shows counts.

9 What makes it different

- **Wristband as interface** — One step at a time, tap to confirm
- **Dependency-aware queue** — Later steps stay locked until ready
- **Voice task entry** — Speak to the band; Whisper transcribes
- **Accounts and roles** — Boss and worker login, password reset, worker status
- **Account-linked band** — Steps go only to that worker’s band
- **Matrix on the wrist** — Browse quadrants, works and steps
- **Offline-tolerant** — Work saved if the band is away; events queued
- **One backend** — App and web share one API and database

10 Dependency locking

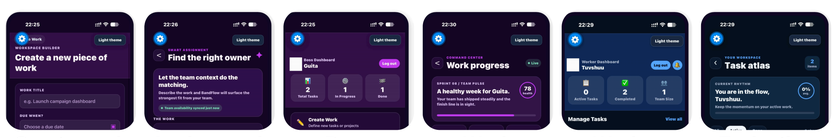
Per task

DONE List ingredients › DONE Draft dish ideas › ACTIVE · ON BAND Cook and taste-test › LOCKED Choose final 3 › LOCKED Cost per dish

- Only the active step is sent to the band.
- A step unlocks when its prerequisite is done; loops are rejected.
- Work can go to several workers; each gets steps on their own band.

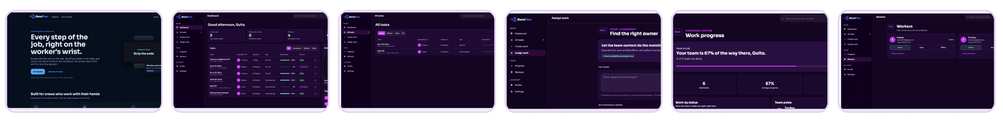
11 Interfaces: app & dashboard

Mobile app — real screenshots



Create work · Assign work · Boss dashboard · Work progress · Worker dashboard · Task atlas

Web dashboard — real screenshots



Landing page · Dashboard · All tasks · Assign work · Work progress · Workers & availability

12 Expected benefits

- Less time spent deciding what to start
- Fewer steps started before prerequisites
- Hands-free steps in busy or noisy work
- Live progress for supervisors

Proposed, not yet measured. Planned test: task completion time with vs. without BandFlow.

BandFlow One sentence: BandFlow turns a big task into ordered steps, delivers them one at a time to a linked wristband, and shows confirmed progress on a shared dashboard.